Why We Changed How "Similarity Between Visits" Is Modeled: A Guide for Clinical Readers

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Purpose. This note explains, without statistical notation where possible, one specific change in how our simulations represent a patient's symptom trajectory over a clinical trial: the assumption about how similar two study visits are to each other, as a function of how far apart in time they are. The original (Hendrickson et al. 2020) simulation code assumed every pair of visits was equally similar, no matter how many weeks apart they fell. The refined process we use now instead assumes similarity fades the further apart two visits are. This note is written for a reader with clinical training but no statistics background, in the style of our more technical `docs/53` and `docs/55` notes, which document two related code-level issues for a statistical audience. It covers the same kind of ground — what changed, why, and what difference it makes — without requiring the reader to follow equations.

1. Summary for the busy reader

The question. In an eight-visit trial (baseline, then weeks 4, 8, 9, 10, 11, 12, 16 and 20), how similar should a patient's symptom score at week 4 be to the same patient's score at week 5, compared with their score at week 20? Both pairs involve "the same patient, two visits," but one pair is a week apart and the other is sixteen weeks apart.

What the original code assumed. Every pair of visits was equally similar — a flat 0.8 correlation between any two visits for the same patient, whether one week or sixteen weeks apart (*quoted*: the project's Methods section states "Within each component the correlation between any two visits is $\rho_{cs} = 0.8$," using Hendrickson's own published value; the package's own default used 0.7 instead). This is called **compound symmetry**: "compound" for multiple visits, "symmetric" in the sense that distance between visits doesn't matter.

What we use now. A value that decays geometrically with the number of weeks between visits — close visits are more alike, distant visits less so (*inspected*, `docs/series/ch10-refined- process.md`; paper 01, Table `tab:spec`). This is called **separable AR(1)**: visits one week apart are correlated at 0.7; visits two weeks apart at $0.7^2 = 0.49$; and so on, multiplying by 0.7 for every additional week of separation.

How different the two assumptions actually are, in numbers. At the trial's most extreme pair of visits (week 4 to week 20, sixteen weeks apart): the old assumption says these two visits are just as correlated as two visits a single week apart (0.8, same as every other pair). The new assumption says they are correlated at $0.7^{16} \approx 0.0033$ — essentially unrelated (*derived*). That is not a subtle difference: the old assumption overstates how predictable a patient's week-20 status is from their week-4 status by roughly two hundred and fifty times.

Why this matters beyond realism. This is not only a question of which assumption is more believable. The old, flat-correlation assumption also puts a hard mathematical ceiling on how strong a biomarker-treatment interaction the simulation can represent at all — regardless of how strong the real effect might be in patients. Under the old assumption, that ceiling is 0.256 (on a scale where 1 would be a perfect, deterministic relationship). Under the new assumption, the ceiling is roughly 0.47 to 0.48 — nearly double (*quoted*, `docs/series/ch10-refined-process.md`: "raises coupling ceiling from 0.256 to 0.480 (Hybrid)"). In plain terms: if the true predictive value of a biomarker like blood pressure for prazosin response were strong — stronger than about one-quarter of the maximum possible — the old simulation code could not represent that at all; it would either silently distort the input correlations to force a mathematically valid solution, or fail outright. The new assumption can represent a substantially stronger, and more clinically plausible, range of biomarker effects.

What does not change. This is strictly about how a patient's *own* symptom score correlates with their *own* score at a different visit. It has nothing to do with how strongly the biomarker itself predicts treatment response (that is a separate parameter, c_{bm}, covered in paper 01's main text) or with how long a drug effect lingers after stopping treatment (carryover, covered in `docs/53`). It is specifically about the shape of "how alike are two visits," a background assumption that the rest of the model is built on top of.

2. Why "every visit equally alike" doesn't match clinical experience

Think about a patient's week-to-week course in a chronic condition like PTSD. Symptom severity drifts over time: a flare near week 4 plausibly still partly explains an elevated score at week 5, but says much less about where that patient will be at week 20, after months of natural fluctuation, life events, and treatment response. Clinical intuition says *recent* visits should predict each other better than *distant* ones.

The old assumption could not represent that at all. It is not that it got the amount of fading wrong — it assumed there was no fading whatsoever. Mathematically, every pair of visits for a given patient, for a given symptom component, shared one single correlation value, however far apart they fell on the calendar. The new assumption adds exactly one new idea: correlation between two visits should depend on how many weeks separate them, multiplying by a fixed "decay factor" (0.7) for each additional week.

3. A worked numeric comparison

The trial design used throughout these simulations (the "Hybrid" design) has visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20 after baseline. The table below shows what each assumption predicts for the correlation between a patient's score at a few representative pairs of visits.

Table 1: Correlation between a patient's own symptom score at two visits, under each assumption (*derived*, decay factor and gaps as stated; the 1-week-apart row uses the package's AR(1) value 0.7 for direct comparability)

Visit pair	Weeks apart	Old assumption (compound symmetry)	New assumption (separable AR(1), decay 0.7/week)
Week 11 vs. week 12	1	0.80	0.70
Week 4 vs. week 8	4	0.80	0.24
Week 4 vs. week 12	8	0.80	0.058
Week 4 vs. week 20	16	0.80	0.0033

Every entry in the "old assumption" column is the same number, regardless of how far apart the visits are — that is what "every visit equally alike" means in practice. The "new assumption" column shrinks steadily as the gap widens, which is the behavior most clinicians would expect from a fluctuating chronic symptom measure.

4. The ceiling consequence, explained without the underlying math

A simulation's correlation assumptions are not free-standing; they have to form a mathematically valid joint distribution (the kind of requirement that, if violated, has no solution — like asking for a triangle whose three sides don't satisfy the triangle inequality). When a biomarker is woven into this correlation structure as a *predictive* factor (the approach this software is built around, explained in paper 01's introduction), there is a maximum strength the biomarker-treatment relationship can have before the overall correlation structure stops being mathematically valid.

That maximum ("ceiling") depends on the same background assumption discussed here — how correlated different visits are with each other. Because the old, flat-correlation assumption packs more raw correlation into every pair of visits (0.8 at every gap, including the largest), it leaves less room for the biomarker-to-response correlation to add before the whole structure becomes invalid. The new, decaying assumption frees up more of that room at longer gaps, since those pairs now contribute much less correlation on their own account. The net effect, worked out precisely elsewhere (`docs/35 -separable-response-covariance.md`, Section 6), is that the maximum representable biomarker effect roughly doubles: from 0.256 to about 0.47-0.48 in the trial design used here.

Concretely: Raskind et al.'s reported association between standing blood pressure and prazosin response in PTSD is the motivating example for this whole research program (paper 01, Introduction). If that association, once translated onto this simulation's correlation scale, turned out to be stronger than 0.256, no version of the old simulation code could have represented it faithfully — not because the effect is implausible, but because the old background assumption about how visits relate to each other left no mathematical room for it. The new assumption removes that artificial restriction for a wide additional range of plausible effect sizes.

5. What we give up by making this change

This change is not free of trade-offs, and a clinically-literate reader deserves to know what they are, not just the benefits.

- **A judgment call, not a proven fact.** Assuming correlation decays in exactly this geometric way, at exactly the same rate for every symptom component, is itself a simplifying assumption — one that can in principle be tested against real longitudinal data and rejected, the way any statistical model can be. Whether this project's three underlying symptom components (natural course, placebo response, and drug response) actually behave this way is not known, because these components are not separately observable in real patients; only their sum is. Adopting this assumption is best read as a considered modeling choice, not an established finding (*inspected*, `docs/35`, Section 5, item 5).
- **Old results no longer reproduce exactly.** Because switching changes the actual numeric correlation values used throughout the simulation (not just at the extremes highlighted above), any simulation result computed under the old assumption will not exactly match one computed under the new one, even holding every other setting fixed. This is expected and intentional, not an error, but it does mean earlier reported numbers under the old assumption are superseded rather than directly comparable.
- **One decay rate for everything.** The new assumption uses a single decay rate (0.7) for all three symptom components alike. It cannot represent, for example, a placebo-belief effect that fades faster than the underlying drug response does. A more flexible version that allows each component its own rate exists in principle but costs the simple, checkable ceiling described above (`docs/35`, Section 8).

6. Where this leaves the simulations

The separable AR(1) assumption (visits decaying in similarity by a fixed factor per week apart) replaces compound symmetry (every pair of visits equally alike) as of the refined process paper 01 recommends. This is one of several changes bundled into that recommendation, alongside the carryover-decay fix described in `docs/53` and the biomarker-coupling fix described in `docs/55`; this note covers only the background visit-to-visit correlation structure, which is a separate and independent piece of the overall refinement.

References

- Project Methods text, `analysis/report/01-dgp-mean-moderation-vs-mvn/report.Rmd`, for the compound-symmetry value as published ($\rho_{cs} = 0.8$) and as implemented in the package default (0.7).

- `docs/series/ch10-refined-process.md`, for the refined process's adoption of separable AR(1) and the quoted ceiling figures.
- `docs/35-separable-response-covariance.md`, the full technical derivation this note simplifies, including the ceiling calculation (Section 6), the advantages-and-disadvantages discussion (Section 5), and the alternative of a per-component decay rate (Section 8).
- `docs/53-carryover-recursion-in-published-code.md` and `docs/55-step-coupling-in-published-code.md`, the companion technical notes on the other two defects in the published code, written for a statistical audience in the same style this note adapts for a clinical one.
- Raskind et al., the prazosin-PTSD blood-pressure association motivating this software (cited in full in paper 01's references).